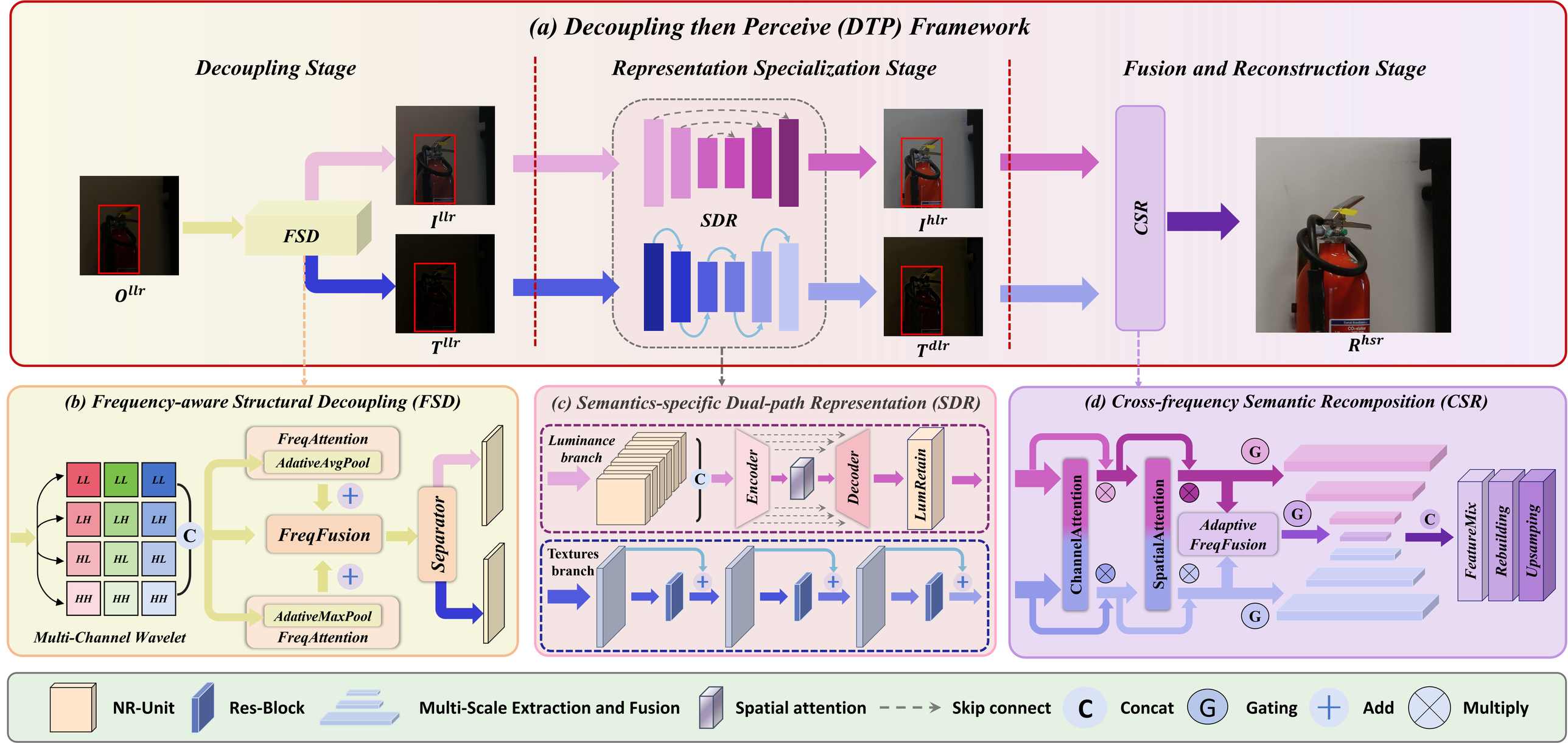
(a) Decoupling then Perceive (DTP) Framework
Decoupling Stage
Representation Specialization Stage
Fusion and Reconstruction Stage
O^{llr}
FSD
I^{llr}
T^{llr}
SDR
I^{hlr}
T^{dlr}
CSR
R^{hsr}
(b) Frequency-aware Structural Decoupling (FSD)
LL
LH
HL
HH
C
Multi-Channel Wavelet
FreqAttention
AdativeAvgPool
FreqFusion
AdativeMaxPool
FreqAttention
Separator
(c) Semantics-specific Dual-path Representation (SDR)
Luminance branch
C
Encoder
Decoder
LumRetain
Textures branch
(d) Cross-frequency Semantic Recomposition (CSR)
ChannelAttention
SpatialAttention
Adaptive FreqFusion
G
C
FeatureMix
Rebuilding
Upsamping
NR-Unit
Res-Block
Multi-Scale Extraction and Fusion
Spatial attention
Skip connect
Concat
Gating
Add
Multiply